Michael Khalfin

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Optimization researcher working on how often we can replace scarce expert judgment with a cheap, biased AI judge, and decide which costly data are worth collecting. I develop practical theory. A good spec is found in conversation with the people who hold it, so I build loops where AI agents draft, cheap checks verify, and experts answer only the questions that change the answer.

EDUCATION

University of Wisconsin–Madison — Madison, WI
Ph.D. student, Industrial and Systems Engineering (optimization) | Research fully funded — Aug 2026 – present

Rice University — Houston, TX
B.S. Operations Research, B.A. Mathematics (GPA 3.76) | Budapest Semesters in Mathematics (GPA 3.85) — Aug 2022 – May 2026
Coursework: Operations Research Models, Linear Optimization, Nonlinear Optimization, Algorithms, Software Development

RESEARCH

Learning an expert panel's preferences from a biased LLM judge — Aug 2026 – present

- Developed a preference elicitation method to align a deployed model to $\geq$95% accuracy by integrating scarce weekly expert feedback with abundant synthetic data from a biased **LLM** judge.
- Proved a **finite-sample accuracy guarantee**, known before any data are collected, and a **cap** on what the LLM adds each week.
- Derived the **optimal split of queries** between LLM and panel, resulting in planning numbers the stakeholder can implement.
- Turned **40+** AI-generated documents into a coherent theory and a deployable algorithm.

Data collection prioritization: a statistical model of uncertainty reduction — Jul 2025 – Oct 2026

- Designed a method that picks valuable data to collect so the numbers a decision rests on are pinned down.
- Encoded the evidence as a mixed-integer program, meeting the industry-standard format **without a prior or noise model**.
- Coded a **6.6M-binary-variable** school-outbreak infection model with on-demand AI-assisted reformulation.
- Devised a four-step workflow to solve any data collection prioritization problem end-to-end, leading to **20 domain formulations** across biology, economics, and engineering.
- Unified worst-case and Bayesian planning in the model: a single dial sets how much to trust prior beliefs, with worst-case as the limit.

Online algorithms (with Sebastian Perez-Salazar), Rice University — Aug 2024 – Dec 2025

- Derived **closed-form competitive ratios** for single-threshold explore-then-exploit rules under immediate, irrevocable decision-making (i.i.d. k-secretary), improving the previous best ratios by **1.3x**.

WORK EXPERIENCE

Sandia National Laboratories — Albuquerque, NM
Operations Research Intern — May 2025 – present

- Directed AI agents to write two **Pyomo-to-Gurobi transpilers**, provably correct for the target model class.
- Created verifiers to test the code: output had to match the reference structurally on 30 reduced instances and on 21 small models that each test one modeling construct; each failed check went back to the agent as a constraint.
- Cut Pyomo mixed-integer linear program model build time at 42M variables from **117 minutes to under 1 minute**.
- Created a domain-specific language to standardize and simplify complex experimental studies (involving optimizing 10,000+ models) and free engineers from writing repetitive and error-prone Python scripts.
- Reconstructed hidden states from partial airline records; decomposed large stochastic models (Benders, progressive hedging), giving a **3x** speedup.

Amazon — Oklahoma City, OK
Area Manager Intern — Jun 2024 – Aug 2024

- Delivered a $5K buffer-storage project, cutting travel time **8%** and saving **over $300K a year**.
- Built a travel map and inventory tracker for a 1.6M sq. ft. site, raising stock accuracy **15%**.

National Laboratory of the Rockies — Denver, CO (Remote)
Supply Chain Automation Intern — Sep 2024 – Oct 2026

- Automated upkeep of the canonical 1,400+ company Li-ion supply-chain database; used by **1,000+** companies for analysis.

PROJECTS

Neural network verification (Rice University, team of 3) — Oct 2025 – Dec 2025

- Encoded a GELU MNIST classifier as an **LP relaxation**; certified its robustness on **600** test digits, with 95% confidence intervals.
- Identified interpretable hidden neurons (Frame, Spine, Belt) that combine to produce each digit's prediction.

LLM elicitation loop on a reMarkable tablet (with Claude, used daily) — Oct 2026 – present

- Built a loop where Claude typesets questions to a tablet, reads the handwritten answers, and refits a preference model each round.
- Applied to a cookbook: translated **8** simple preference choices into a linear model that evaluated all **20** recipes to isolate the **4** best.

AWARDS, PUBLICATIONS, AND SKILLS

Awards: CMOR Franco Award (2025). **1st** in HackRice 13 (2023), top 20% in SICK Solution Hackathon (2023), **1st** in UPenn Hackathon Github Track (2023).

Publications (first author):

- "Vehicle-to-Everything Communication Using a Roadside Unit for Over-the-Horizon Object Awareness." *IEOM* 2023 (oral; winner).
- "A Computational Model of the Trans-Synaptic Spread of Pathogenic Tau in Early Alzheimer's Disease." *IMPULSE* 2021.

Methods: LP/MILP, decomposition, stochastic programming, imprecise probability, experimental design, preference elicitation, LLM-as-judge evals. **Tools:** Python, Pyomo, Gurobi, Claude Code, Codex, C++, Lean, Rocq.